

TAKSH VADALIYA

AI SystemsEngineer • Multi-AgentOrchestration • RAG Pipelines

vadaliyataksh06@gmail.com • +91 6353774726 • github.com/Taksh-v • taksh-v.github.io • linkedin.com/in/taksh-vadaliya

SUMMARY

Self-taught AI systems engineer specializing in multi-agent orchestration and RAG pipelines. Expert in designing heterogeneous local LLM architectures with cognitive task routing, 4-metric custom evaluation frameworks, and agentic swarm systems for fintech and enterprise knowledge environments. Obsessed with autonomous systems that plan, reason, and act.

Currently Building Assest, an agentic company brain, in active incubation at AGF (Atmanirbhar Gujarat Fellowship)

TECHNICAL SKILLS

Tools & Framework

AI & LLMs:	Local LLM Architectures, Multi-Agent Orchestration, Agentic Swarm Systems
RAG & Storage:	Hybrid Retrieval Pipelines, Embeddings, Vector & Graph Databases
Backend & Infra:	Python, FastAPI, REST APIs, Systems Programming (C)
Evaluation:	Custom tools & Frameworks, Quality Gates, Data Auditing Pipelines

PROJECTS Experience

Assest — Agentic Company Brain

web-kappa-eight-88.vercel.app

Python, FastAPI, Qdrant, Graph DB, Mastra • Remote (In Development)

- Engineered a high-precision knowledge engine that unifies fragmented organization data from startup-usable applications into a queryable asset. [Latency 1.2s, Retrieval Accuracy 6.3, response time 4.7s]
- Architected a hybrid storage layer merging vector embeddings with graph databases for real-time relationship mapping at scale.
- Deployed a fully functional working prototype on Vercel backed by optimized custom agentic memory pipeline systems.

Qual Finance Analysis System

github.com/Taksh-v/Qual

Python, LangChain, Qdrant, RAG • Open-source

- Built a modular financial intelligence RAG pipeline structured for end-to-end ingestion, chunking, and embedding generation.
- Optimized a custom 4-metric system tracking hallucination, completeness, relevancy, and faithfulness to enforce retrieval integrity.
- Validated pipeline processing pipelines with a dedicated data auditing layer, securing an aggregate verification score of 0.704. (baseline : ~0.5), exceeding industry threshold of 0.65

DDAS — Data Duplication Alert System

github.com/Taksh-v/The-DDAS

C, Systems Programming, File I/O • Systems Development

- Engineered a low-level systems file utility in C to locate duplicate system assets across deeply nested storage trees.
- Implemented custom low-level file-hashing routines and optimized hash deduplication to index complex hierarchies rapidly.
- Managed manual memory buffers and explicit file descriptor states to prevent resource leaks during intense I/O loops.

ACHIEVEMENTS & RECOGNITION

- GTU Ventures / Gujarat Fellowship:** Founded Assest as an independent startup; pitched for institutional incubation and currently navigating the startup fellowship and regional SSIP pipelines.
- Devang Mehta Hackathon 2025 (GTU Flagship):** Led the system building track as team lead; designed, engineered, and demonstrated an end-to-end operational platform within strict competitive timeboxes.
- Smart India Hackathon 2024:** Qualified for the competitive State Round of this ministry-backed national innovation event among thousands of teams across the country.

EDUCATION & CREDENTIALS

B.Sc. Information Technology — GTU-ITR (2023-2026)

CGPA: 8.5

Certifications: Harvard/edX CS50 Python • Mastercard Cybersecurity • Cisco Endpoint Security • **Anthropic Claude101**

INTERESTS & COMMUNITY

- Distance Runner:** Trains consistently; applies the same iterative improvement mindset to long-distance athletic goals as to complex backend systems architecture problems.
- AI/ML Community Contributor:** Actively building a technical voice on Reddit and LinkedIn focused on agentic frameworks, production RAG pipelines, and open-source local LLM deployments.